VIZSGAREMEK

Készítették:

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Konzulens:

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Miskolc

2026.

Miskolci SZC Kandó Kálmán Informatikai Technikum

Miskolci Szakképzési Centrum

Informatikai rendszer- és alkalmazás-üzemeltető technikus

Tesztelési Dokumentáció

Katona Dominik - Tóth Ádám

2025-2026

Tartalom

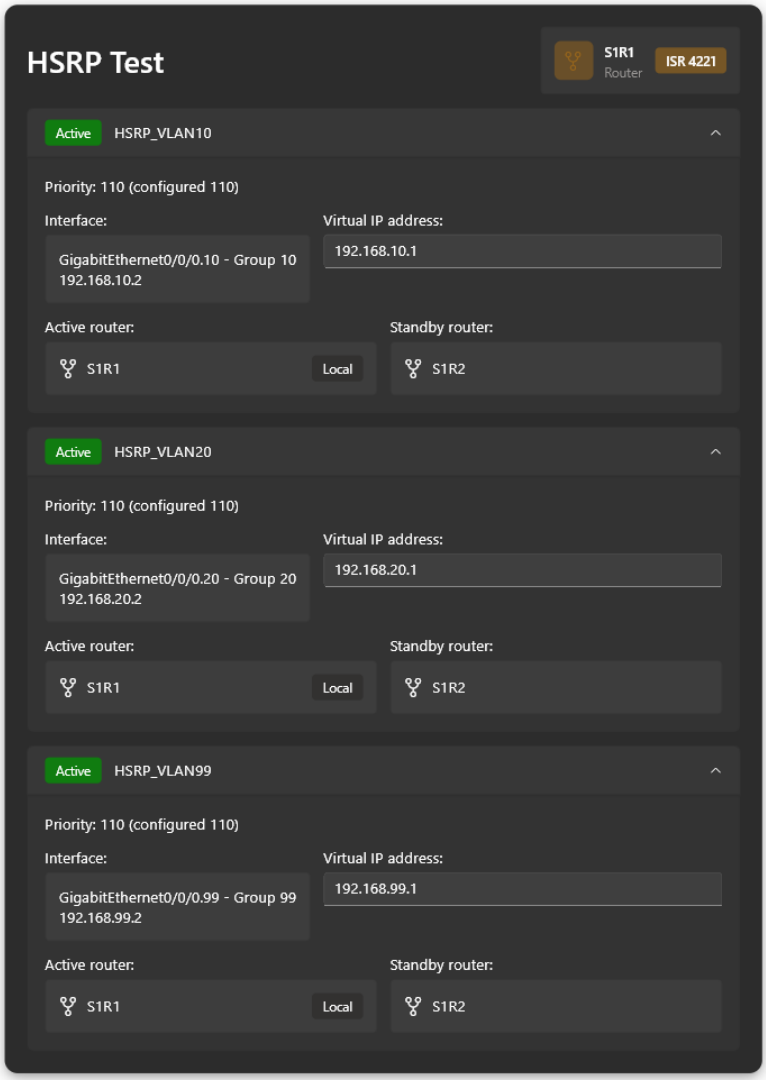
1 Iroda (Komplex) telephely 4
1.1 Routerek 4
1.2 Switchek 8
1.3 Tűzfal 10
2 Szerverek telephely 11
2.1 Router 11
2.2 Windows szerver 14
2.2.1 DHCP 14
2.2.2 Samba 15
2.2.3 AD 15
2.2.4 DNS 16
2.2.5 RADIUS 16
2.3 Linux szerver 16
2.3.1 Samba 17
2.3.2 Weboldalak 18
2.3.3 Adatbázis & API 19
3 Raktár (Wifi) telephely 19
3.1 Router 19
3.2 AP 21
4 Project management 21

1 Iroda (Komplex) telephely

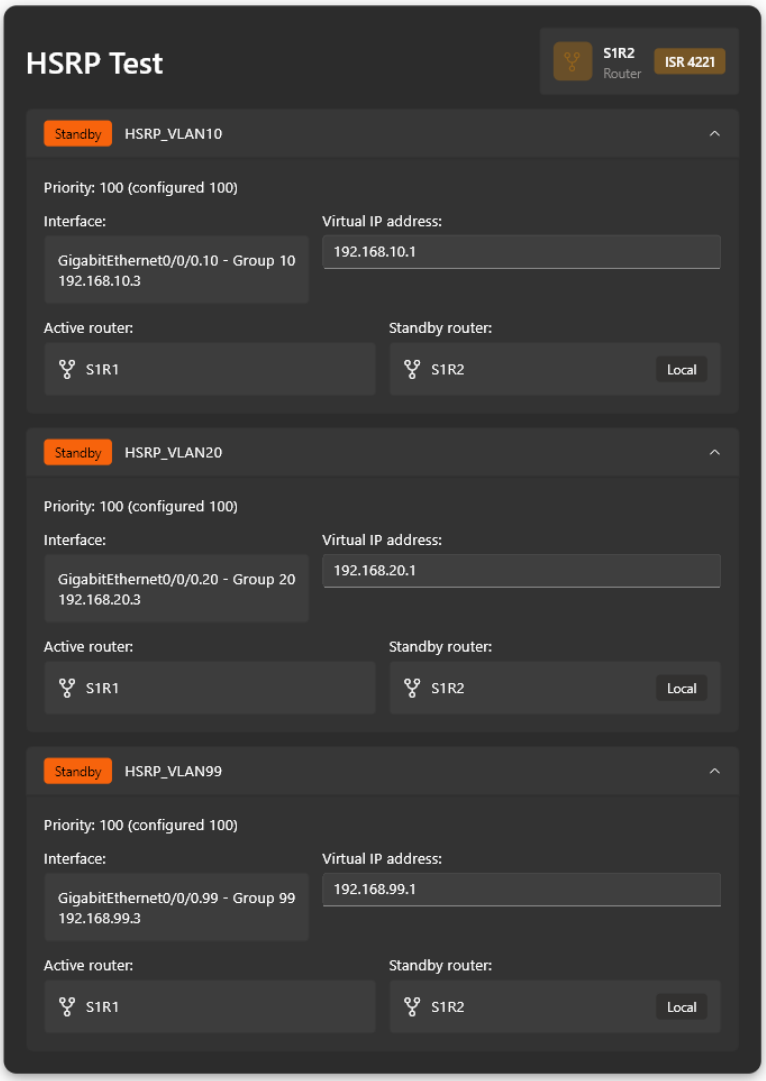
1.1 Routerek

HSRP

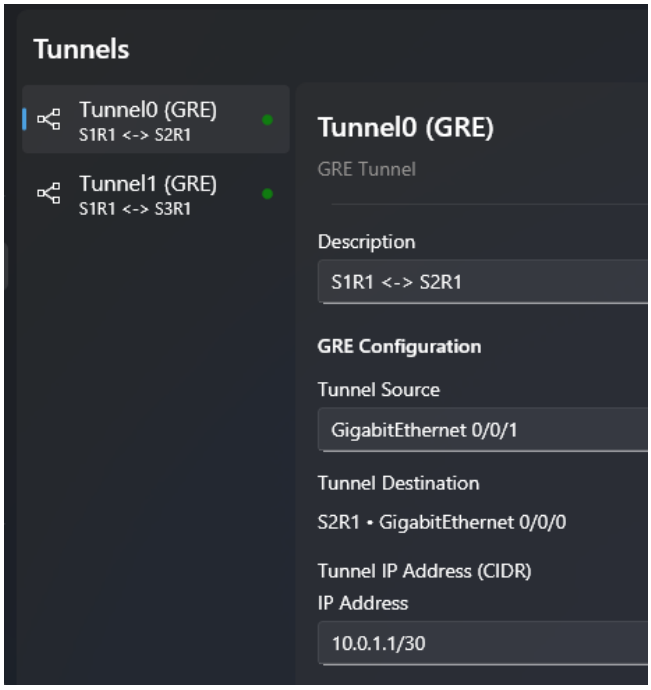
Router 1, Active

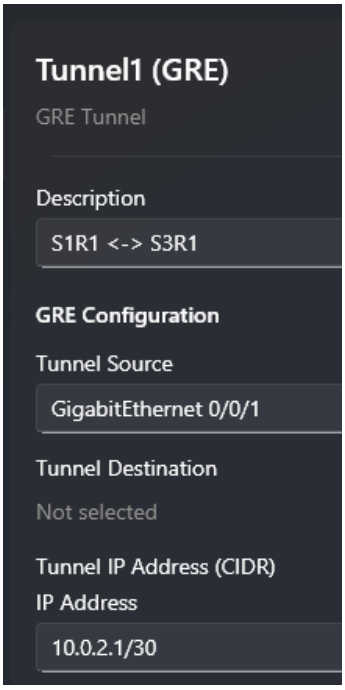


Router 2, Standby



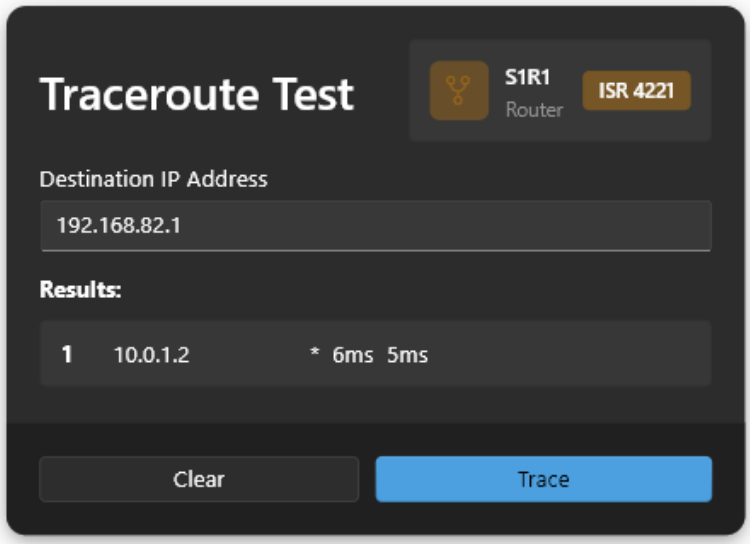
Tunnel



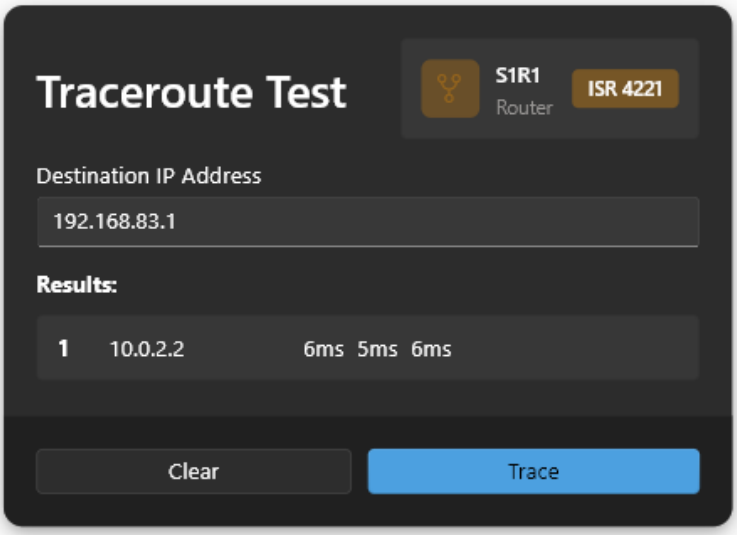


Traceroute

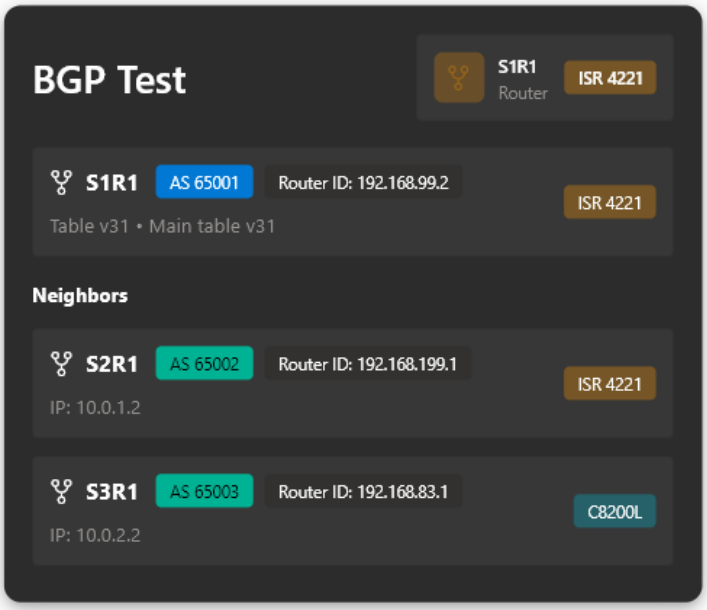
Telephely 2



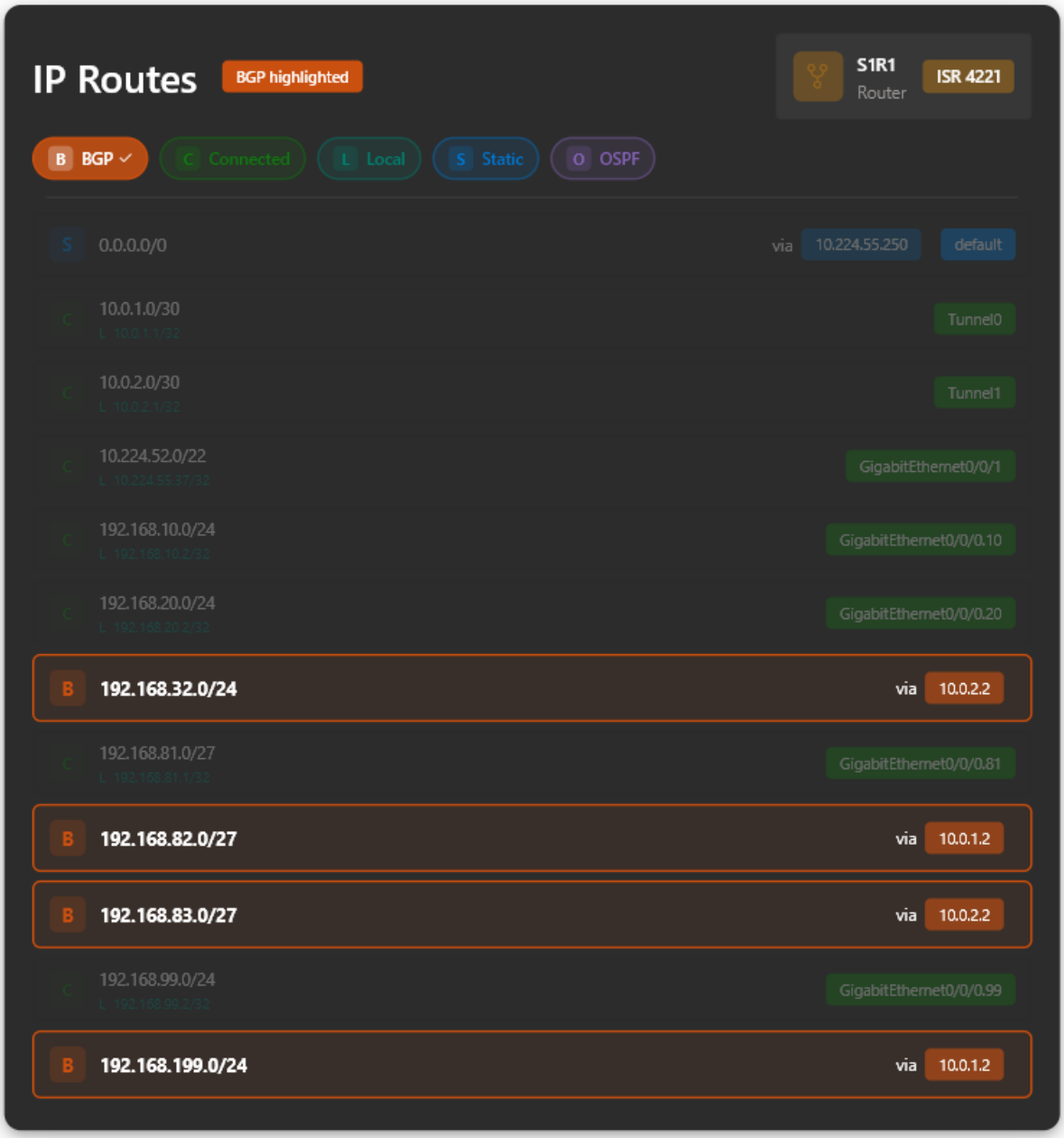
Telephely 3



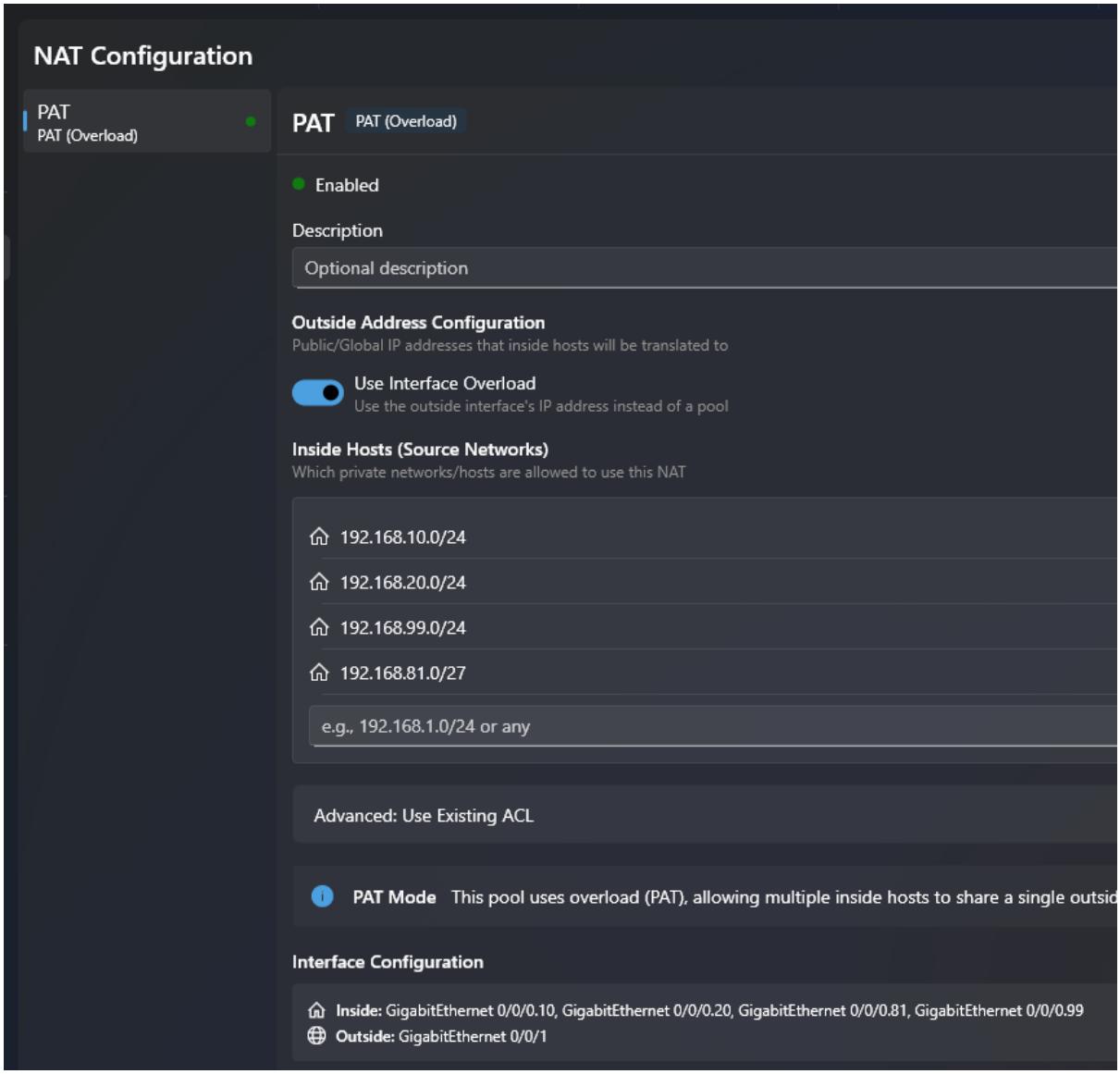
BGP


BGP Test
S1R1
Router
ISR 4221
S1R1
AS 65001
Router ID: 192.168.99.2
ISR 4221
Table v31 • Main table v31
Neighbors
S2R1
AS 65002
Router ID: 192.168.199.1
ISR 4221
IP: 10.0.1.2
S3R1
AS 65003
Router ID: 192.168.83.1
C8200L
IP: 10.0.2.2

IP route

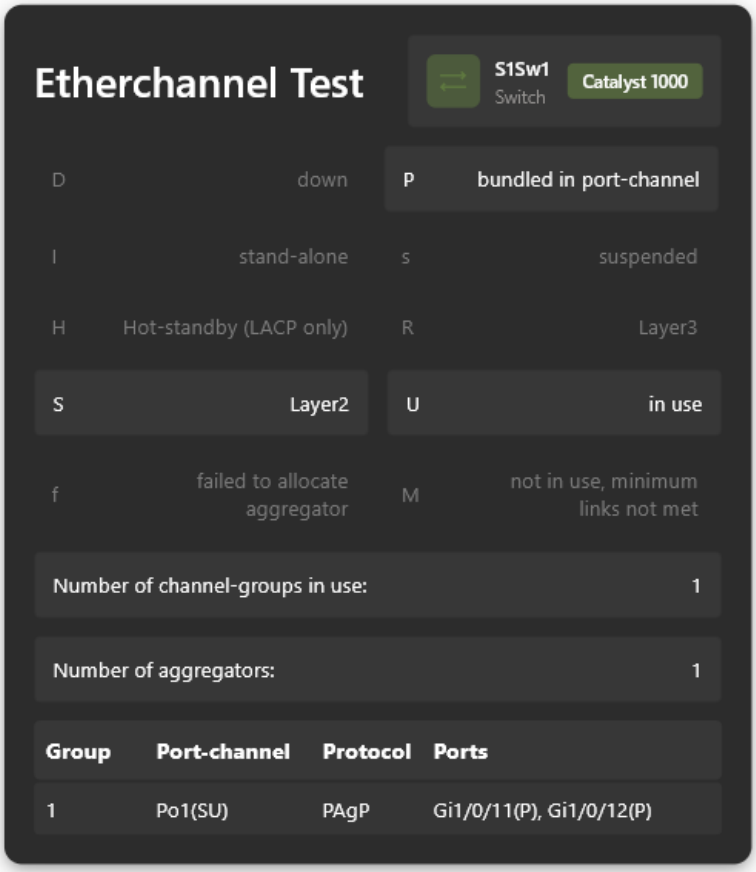

IP Routes
BGP highlighted
S1R1
Router
ISR 4221
B BGP
C Connected
L Local
S Static
O OSPF
S 0.0.0.0/0 via 10.224.55.250 default
C 10.0.1.0/30 Tunnel0
C 10.0.2.0/30 Tunnel1
C 10.224.52.0/22 GigabitEthernet0/0/1
C 192.168.10.0/24 GigabitEthernet0/0/0.10
C 192.168.20.0/24 GigabitEthernet0/0/0.20
B 192.168.32.0/24 via 10.0.2.2
C 192.168.81.0/27 GigabitEthernet0/0/0.81
B 192.168.82.0/27 via 10.0.1.2
B 192.168.83.0/27 via 10.0.2.2
C 192.168.99.0/24 GigabitEthernet0/0/0.99
B 192.168.199.0/24 via 10.0.1.2

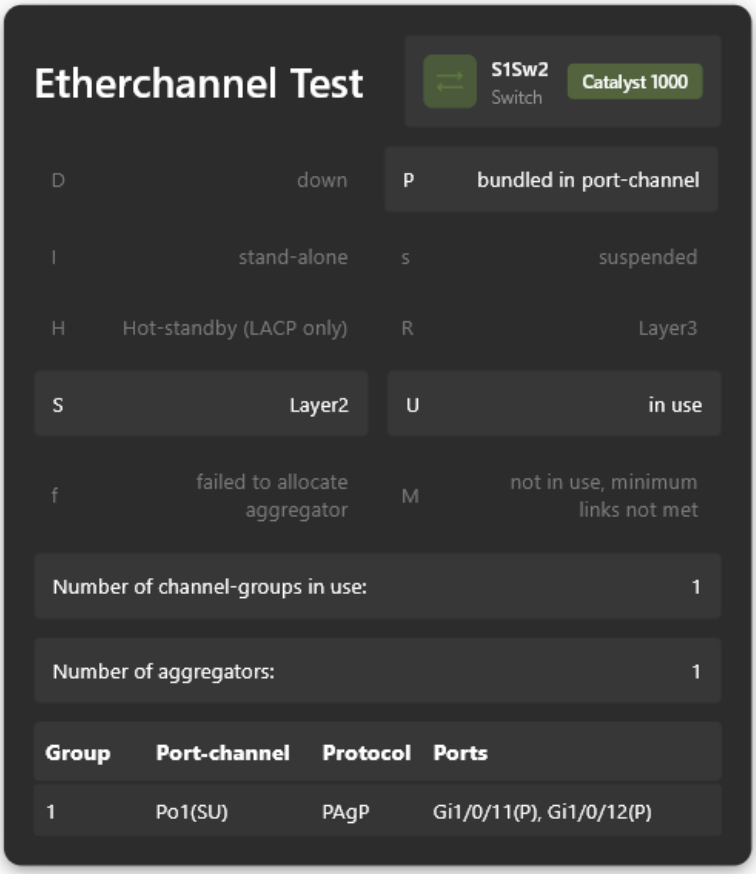
PAT


NAT Configuration
PAT
PAT (Overload)
PAT
PAT (Overload)
Enabled
Description
Optional description
Outside Address Configuration
Public/Global IP addresses that inside hosts will be translated to
Use Interface Overload
Use the outside interface's IP address instead of a pool
Inside Hosts (Source Networks)
Which private networks/hosts are allowed to use this NAT
192.168.10.0/24
192.168.20.0/24
192.168.99.0/24
192.168.81.0/27
e.g., 192.168.1.0/24 or any
Advanced: Use Existing ACL
PAT Mode This pool uses overload (PAT), allowing multiple inside hosts to share a single outsid
Interface Configuration
Inside: GigabitEthernet 0/0/0.10, GigabitEthernet 0/0/0.20, GigabitEthernet 0/0/0.81, GigabitEthernet 0/0/0.99
Outside: GigabitEthernet 0/0/1

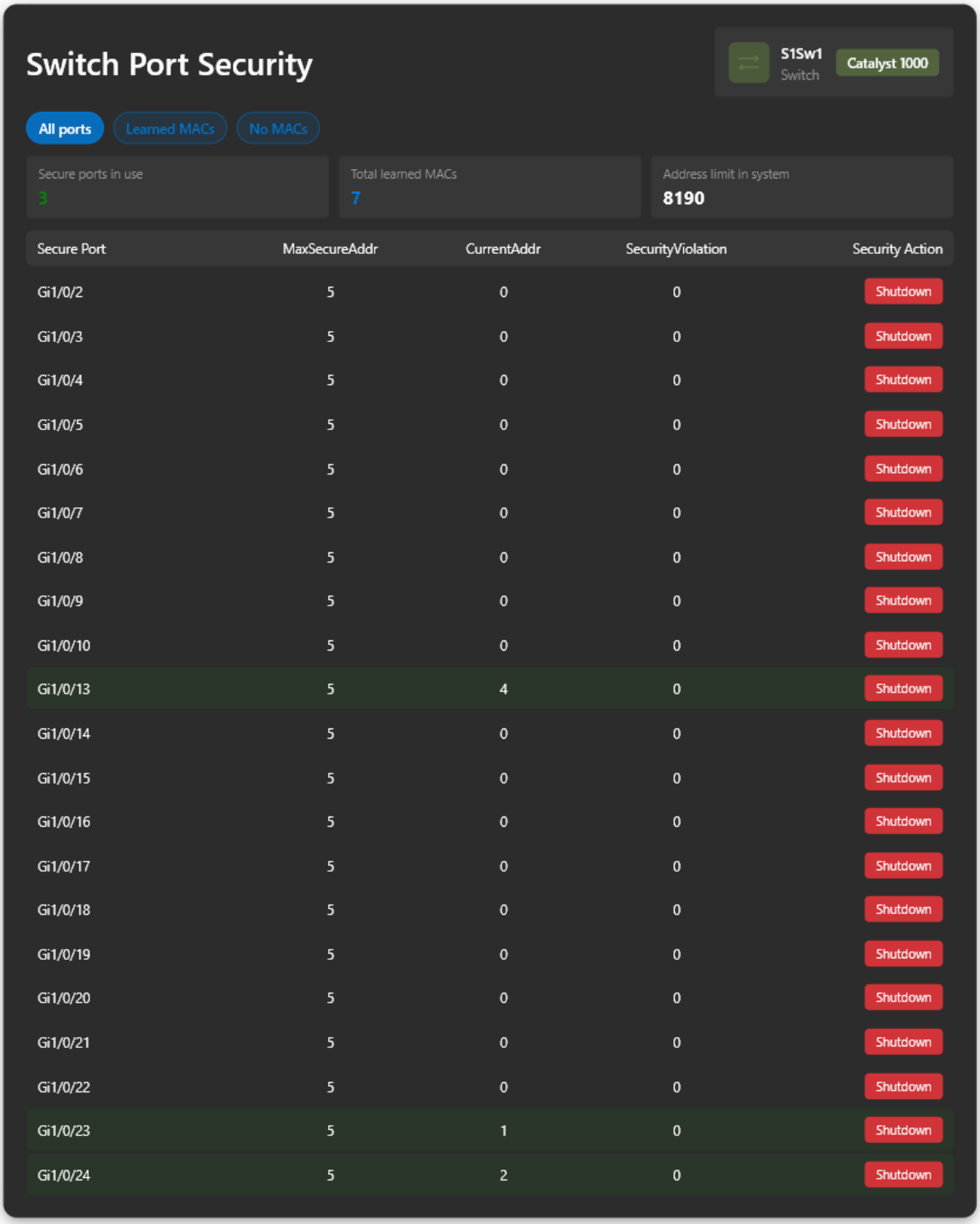
1.2 Switchek

Etherchannel





Portbiztonság



Switch Port Security

S1Sw1 Switch Catalyst 1000

All ports | Learned MACs | No MACs

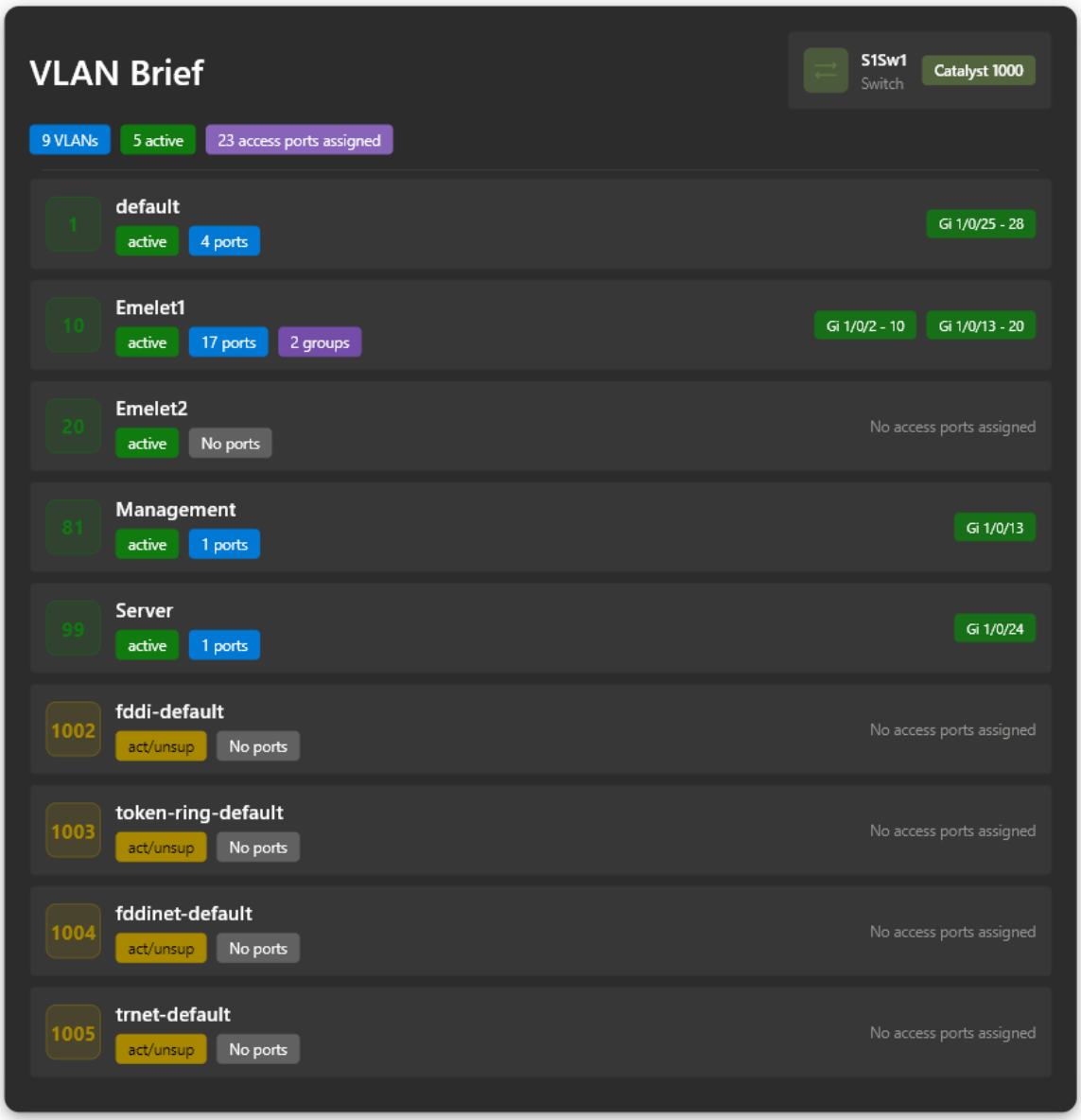
Secure ports in use: 3

Total learned MACs: 7

Address limit in system: 8190

Secure Port	MaxSecureAddr	CurrentAddr	SecurityViolation	Security Action
Gi1/0/2	5	0	0	Shutdown
Gi1/0/3	5	0	0	Shutdown
Gi1/0/4	5	0	0	Shutdown
Gi1/0/5	5	0	0	Shutdown
Gi1/0/6	5	0	0	Shutdown
Gi1/0/7	5	0	0	Shutdown
Gi1/0/8	5	0	0	Shutdown
Gi1/0/9	5	0	0	Shutdown
Gi1/0/10	5	0	0	Shutdown
Gi1/0/13	5	4	0	Shutdown
Gi1/0/14	5	0	0	Shutdown
Gi1/0/15	5	0	0	Shutdown
Gi1/0/16	5	0	0	Shutdown
Gi1/0/17	5	0	0	Shutdown
Gi1/0/18	5	0	0	Shutdown
Gi1/0/19	5	0	0	Shutdown
Gi1/0/20	5	0	0	Shutdown
Gi1/0/21	5	0	0	Shutdown
Gi1/0/22	5	0	0	Shutdown
Gi1/0/23	5	1	0	Shutdown
Gi1/0/24	5	2	0	Shutdown

VLANok



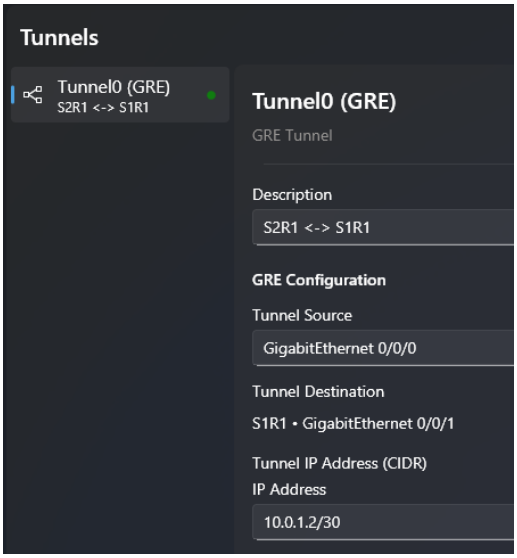
1.3 Tűzfal

?

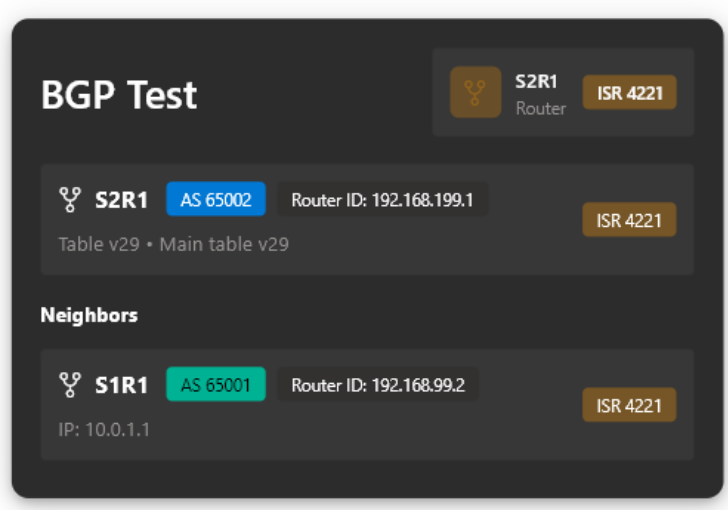
2 Szerverek telephely

2.1 Router

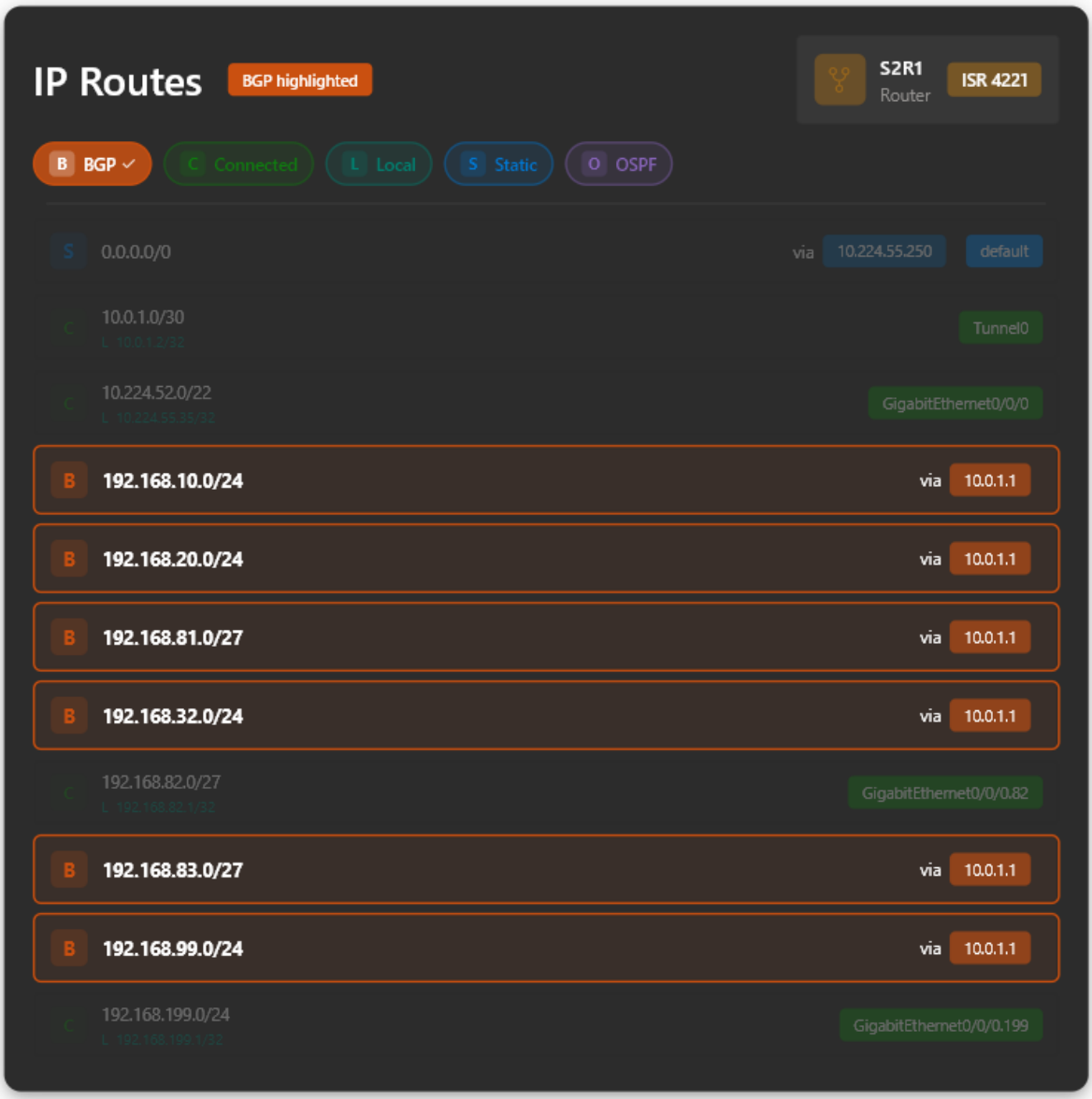
Tunnel



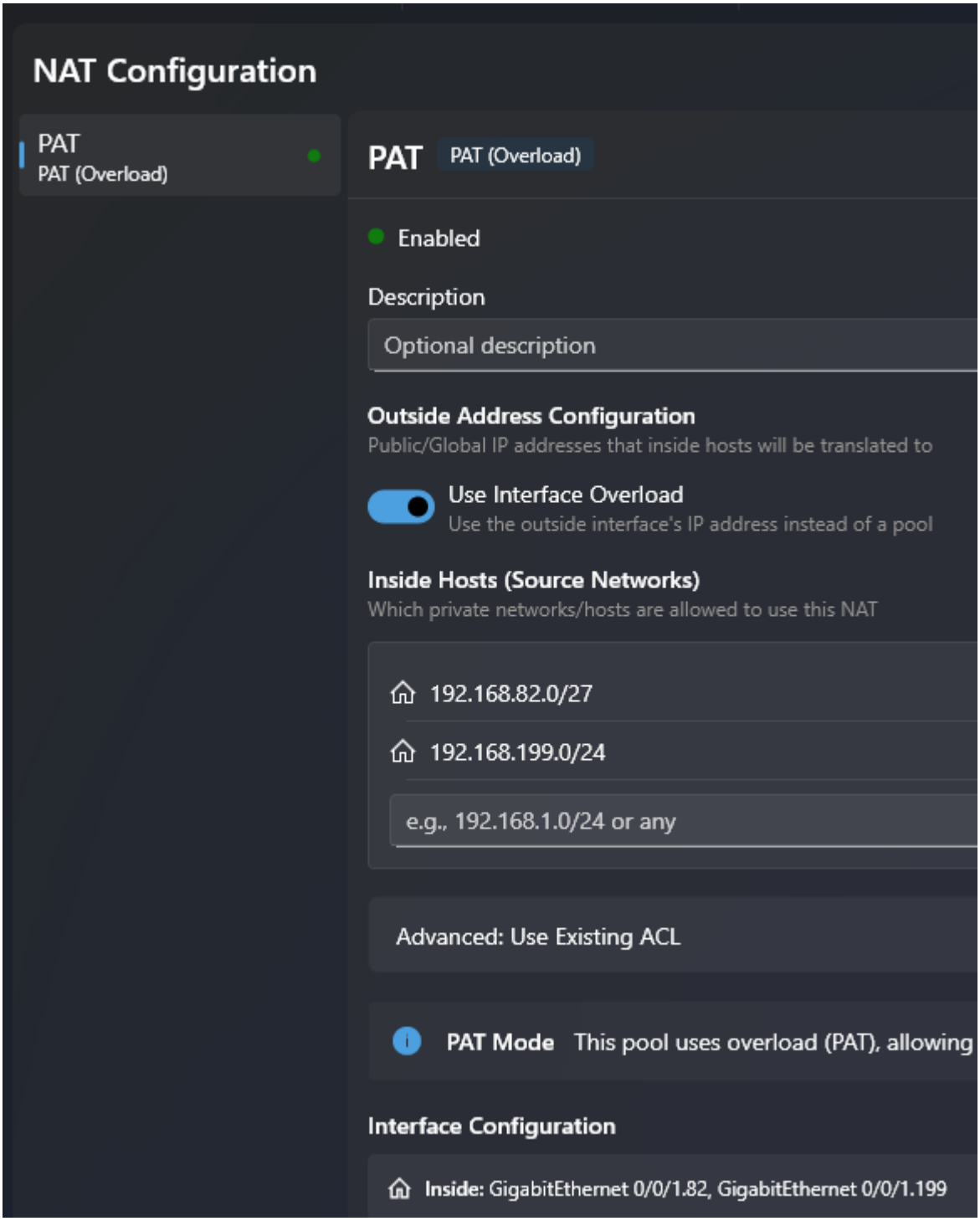
BGP



IP route


IP Routes
BGP highlighted
S2R1
Router
ISR 4221
B BGP
C Connected
L Local
S Static
O OSPF
S 0.0.0.0/0 via 10.224.55.250 default
C 10.0.1.0/30 Tunnel0
C 10.224.52.0/22 GigabitEthernet0/0/0
B 192.168.10.0/24 via 10.0.1.1
B 192.168.20.0/24 via 10.0.1.1
B 192.168.81.0/27 via 10.0.1.1
B 192.168.32.0/24 via 10.0.1.1
C 192.168.82.0/27 GigabitEthernet0/0/0.82
B 192.168.83.0/27 via 10.0.1.1
B 192.168.99.0/24 via 10.0.1.1
C 192.168.199.0/24 GigabitEthernet0/0/0.199

PAT

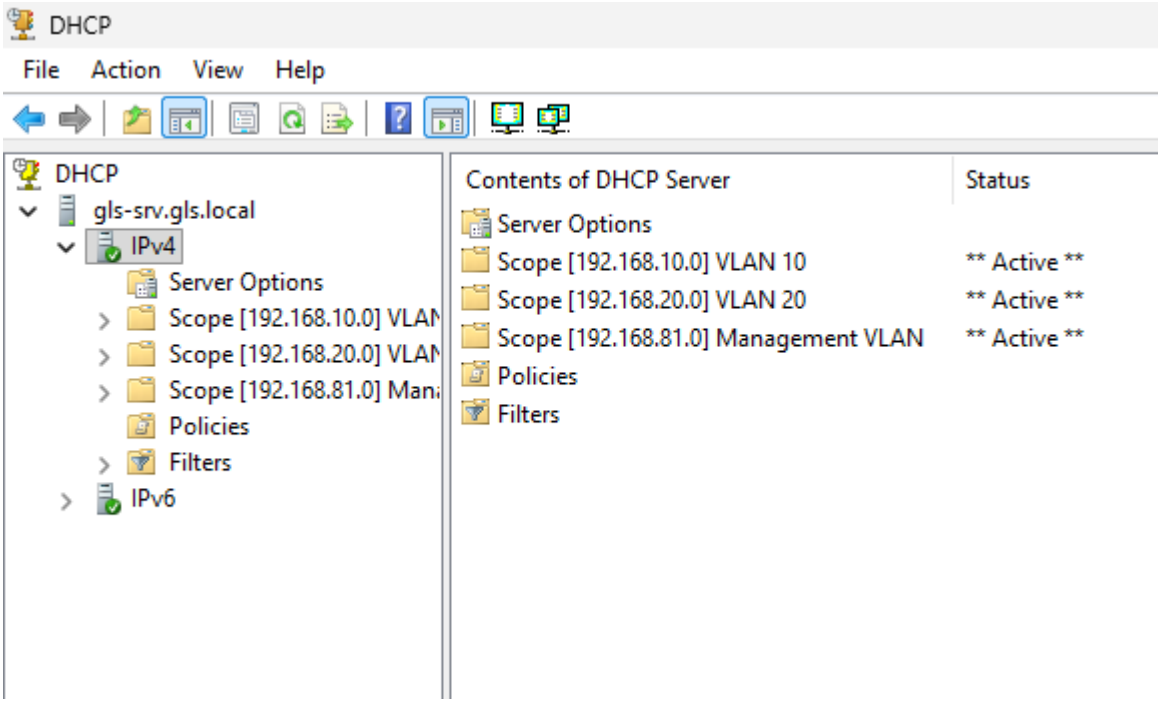

NAT Configuration
PAT
PAT (Overload)
PAT
PAT (Overload)
Enabled
Description
Optional description
Outside Address Configuration
Public/Global IP addresses that inside hosts will be translated to
Use Interface Overload
Use the outside interface's IP address instead of a pool
Inside Hosts (Source Networks)
Which private networks/hosts are allowed to use this NAT
192.168.82.0/27
192.168.199.0/24
e.g., 192.168.1.0/24 or any
Advanced: Use Existing ACL
PAT Mode This pool uses overload (PAT), allowing
Interface Configuration
Inside: GigabitEthernet 0/0/1.82, GigabitEthernet 0/0/1.199

2.2 Windows szerver

Windows server 2025

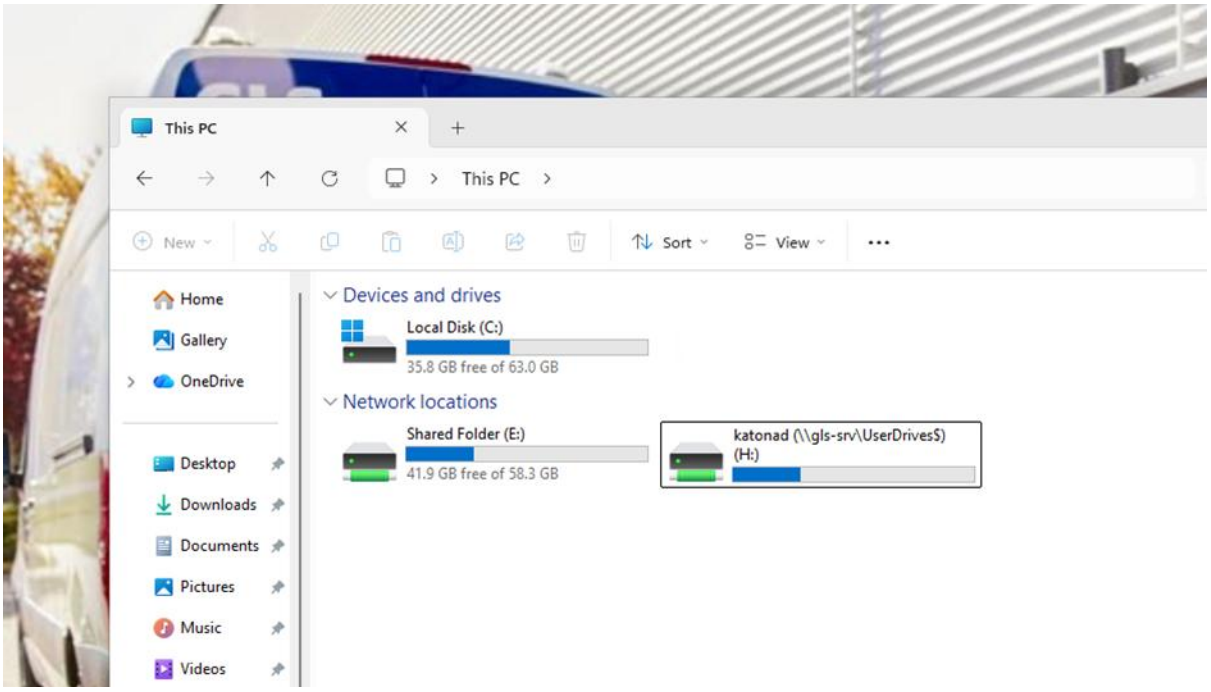
2.2.1 DHCP

VLAN 10, 20, 80nak



2.2.2 Samba

Minden felhasználó eléri a céges közös mappát (Shared Folder), és mindenki kap egy saját HOME mappát Active Directory drive mapből



2.2.3 AD

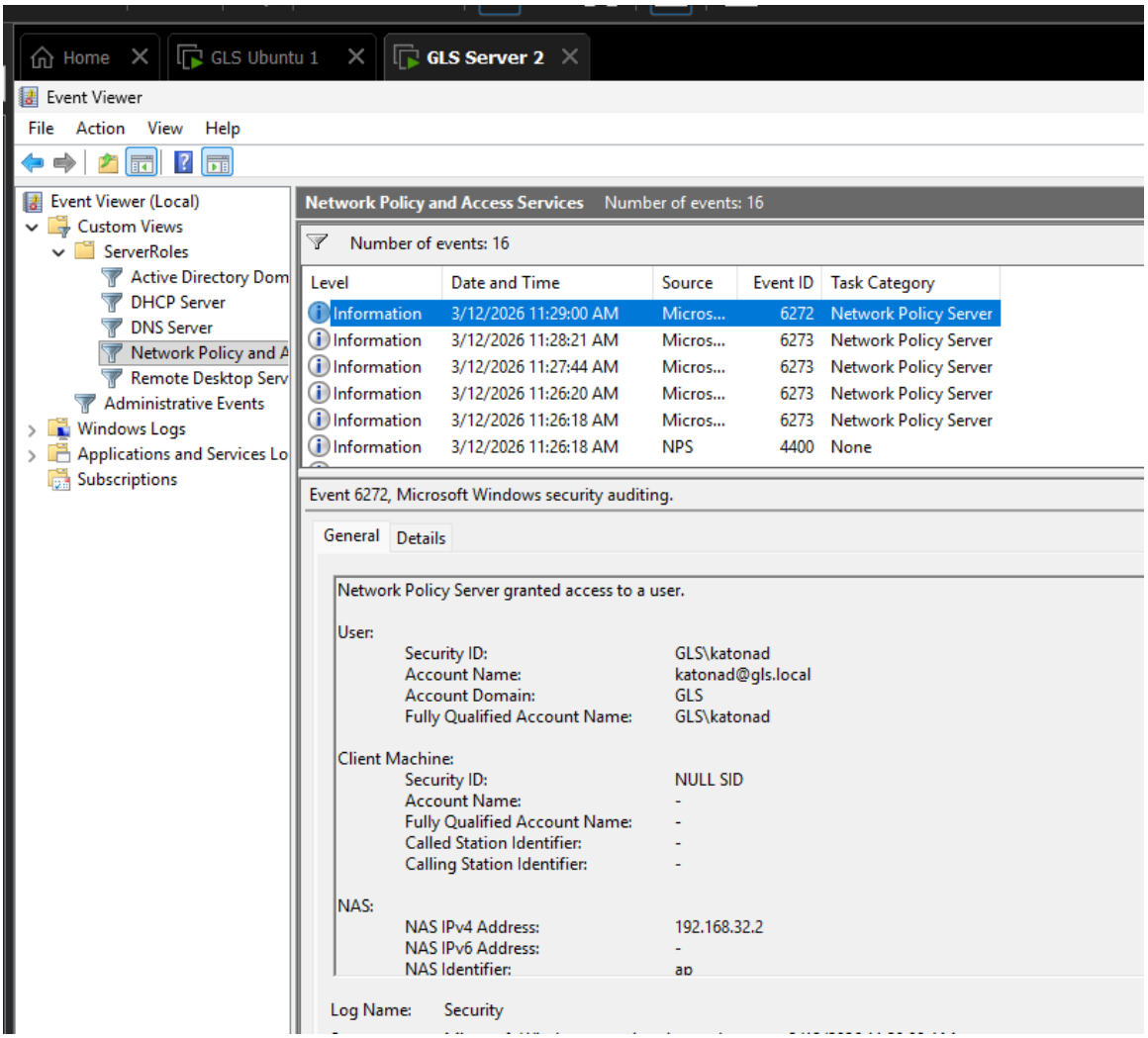
Csoportok, felhasználók

Name	Type	Description
Call Center	Security Group - Domain Local	
GLS-PC-1	Computer	
Katona Dominik	User	
Tóth Ádám	User	
WIN-PC-01	Computer	

2.2.4 DNS

```
gls@gls-ubuntu-1:~$ nslookup api.pacex.hu
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   api.pacex.hu
Address: 192.168.199.253
```

2.2.5 RADIUS



2.3 Linux szerver

Ubuntu 24.04

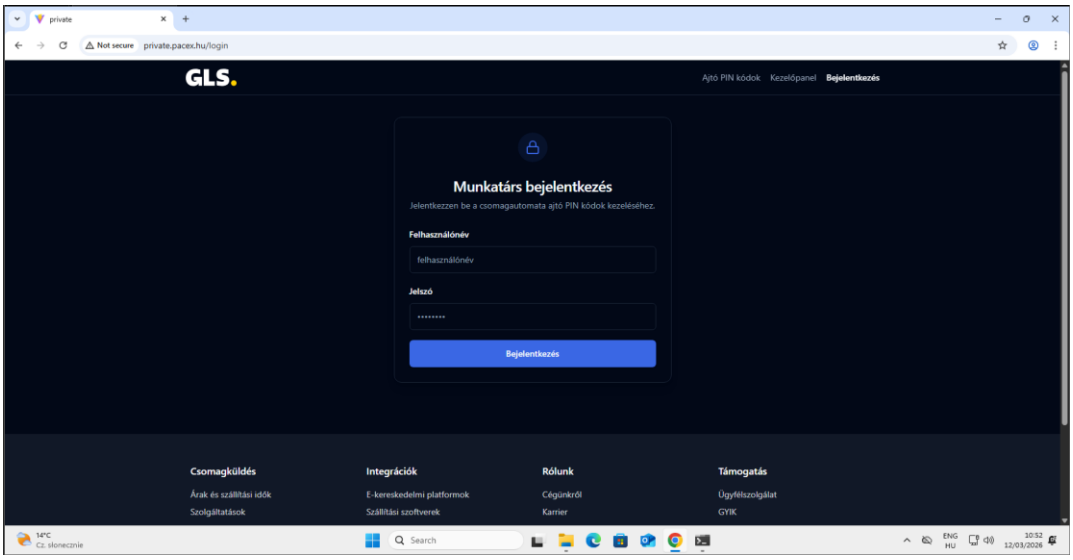
2.3.1 Samba

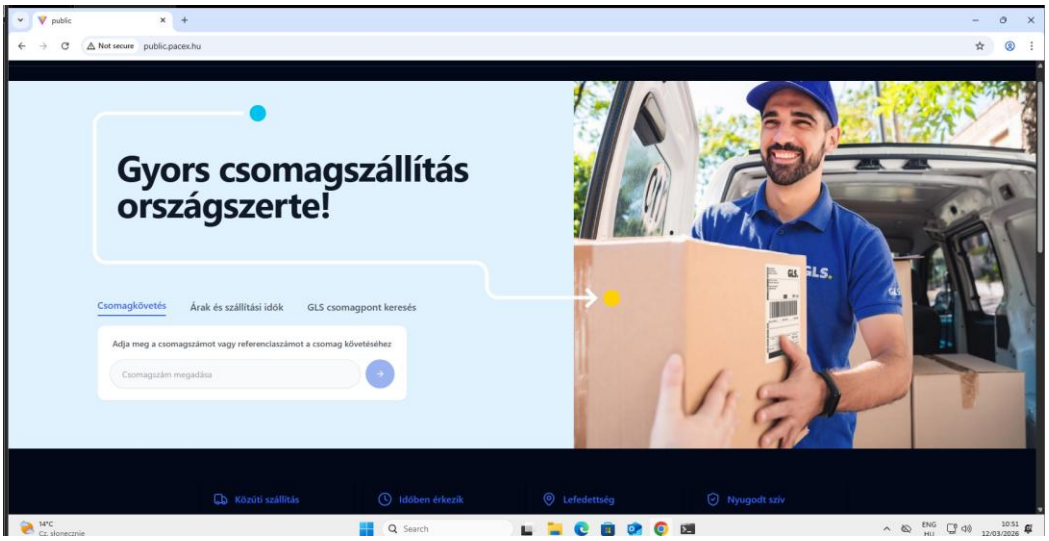
Egész céges közös meghajtó

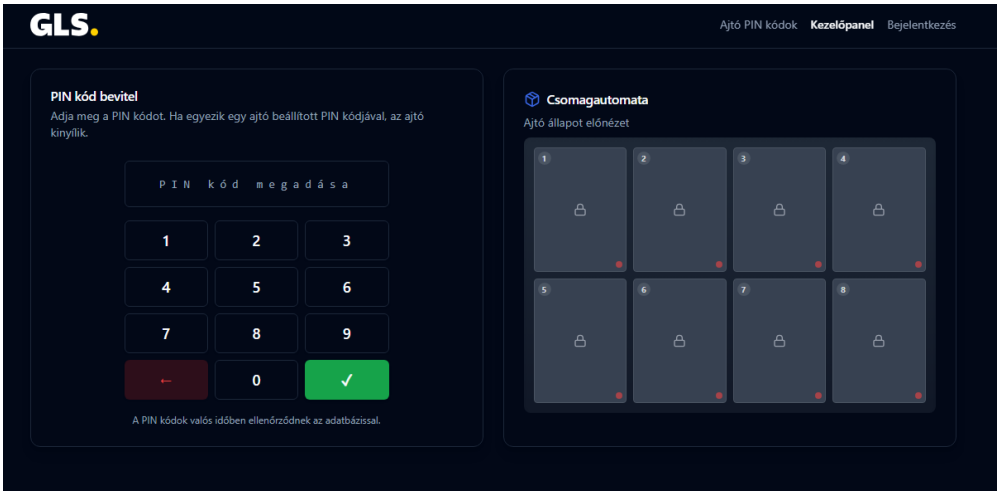
```
[PublicShare]
        guest ok = Yes
        path = /srv/samba/public_share
        read only = No


[ReadOnly]
        create mask = 0644
        guest ok = Yes
        path = /srv/samba/ro_share
        write list = gls @gls
gls@gls-ubuntu-1:~$
```

2.3.2 Weboldalak

Private, Public







2.3.3 Adatbázis & API

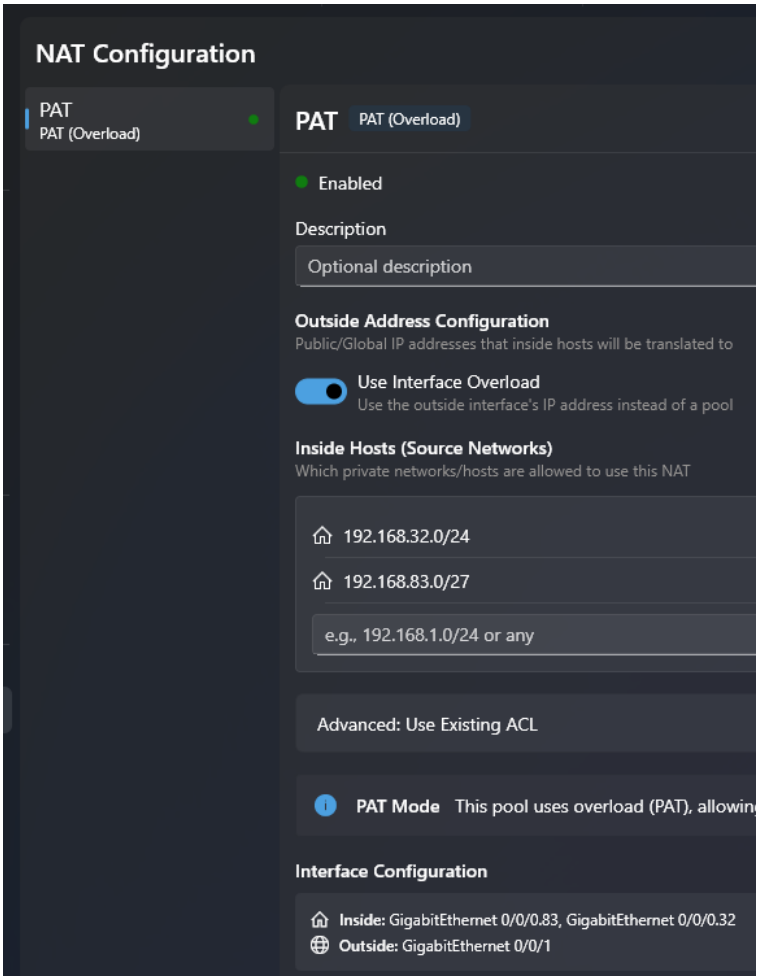
PostgreSQL, python api, uvicorn

```
gls@gls-ubuntu-1:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
     Active: active (exited) since Thu 2026-03-12 07:20:45 CET; 3h 37min ago
    Process: 1802 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1802 (code=exited, status=0/SUCCESS)
        CPU: 10ms

Mar 12 07:20:45 gls-ubuntu-1 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Mar 12 07:20:45 gls-ubuntu-1 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
gls@gls-ubuntu-1:~$
```

3 Raktár (Wifi) telephely

3.1 Router

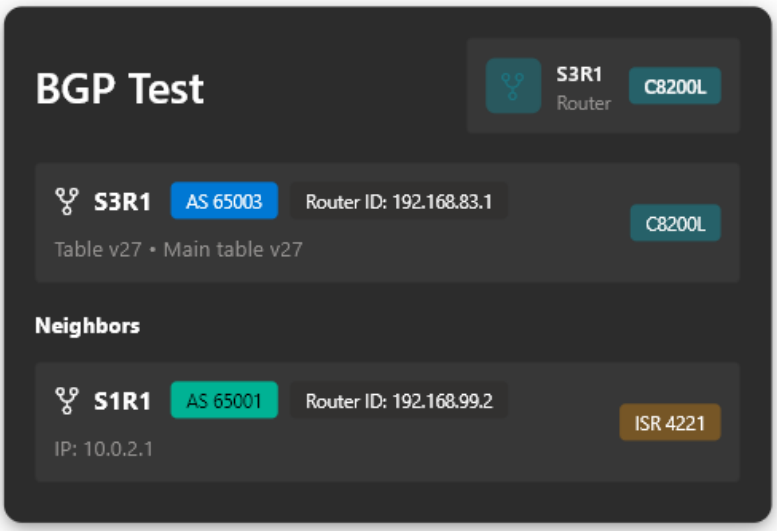
PAT



Tunnel

Tunnels
Tunnel0 (GRE)
S3R1 <-> S1R1
Tunnel0 (GRE)
GRE Tunnel
Description
S3R1 <-> S1R1
GRE Configuration
Tunnel Source
GigabitEthernet 0/0/1
Tunnel Destination
S1R1 • GigabitEthernet 0/0/1
Tunnel IP Address (CIDR)
IP Address
10.0.2.2/30

BGP


BGP Test
S3R1
Router
C8200L
S3R1
AS 65003
Router ID: 192.168.83.1
C8200L
Table v27 • Main table v27
Neighbors
S1R1
AS 65001
Router ID: 192.168.99.2
ISR 4221
IP: 10.0.2.1

3.2 AP

Aironet 1100 eszköz

Hirdeti a hálózatot, de nem lehet bejelentkezni, szerintünk az eszköz hibás (a project végére debug és ssh sem működött)

Hogyha SSH-n keresztül indítunk tesztet a RADIUS szerver felé, akkor működik (2.2.5 RADIUS)

4 Project management

Github